

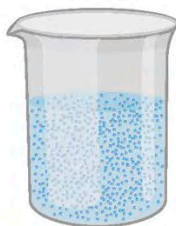
Mixtures

A mixture is a combination of substances that can be separated by physical means, such as filtering. It is not the same as a compound, where the ingredients are connected by bonds and can only be separated using a chemical reaction. Mixtures can be classified as solutions, colloids, and suspensions.



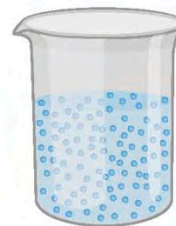
Solution

In this mixture, a substance is completely and evenly mixed, or dissolved, into another substance. Seawater is a solution.



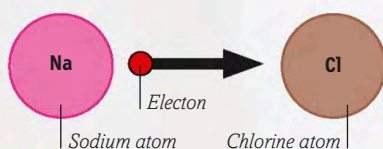
Colloid

This mixture contains unevenly spread particles and clusters that are too small to see. Milk is a colloid.



Suspension

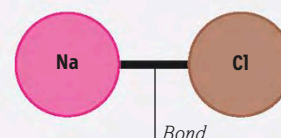
This type of mixture consists of large particles of one substance floating in another substance. Muddy water is a suspension.



1. A sodium atom donates one electron to a chlorine atom. This gives both atoms full outer electron shells.



2. These are now charged atoms known as ions. The sodium ion has a positive charge and the chlorine ion has a negative charge.



3. Sodium is attracted to – and forms a bond with – chlorine, forming a molecule of the compound sodium chloride.

Forming compounds

There are two kinds of bonds formed between elements during a chemical reaction. In an ionic bond, such as in sodium chloride (above), one atom gives away its electron(s) and another accepts them. This results in each having full outer electron shells. The other type is called covalent bonding. In this, atoms sit together and share their electrons so they both have full outer shells.

As lithium burns in air, it becomes lithium oxide.



Reactions in the real world

Chemical reactions happen all around us. There are reactions when we cook, take medication, or breathe. The image above shows a rusty iron ship. Over time, the element iron develops this red, flaky layer when it reacts with oxygen present in water or air to form the compound iron oxide – more commonly known as rust.